

Spatial Analysis of Cell Populations from Fluorescence Microscopy

Literature review for the fly_ovaries_processing_pipeline project

PubMed search conducted March 2026

This review covers published work most relevant to segmenting cell populations from 3D fluorescence microscopy and characterizing their spatial distribution and connectivity. The focus is on: (1) Ripley's K and related spatial statistics, (2) cell contact graph and network analysis, (3) 3D segmentation + spatial phenotyping pipelines, and (4) statistical controls to avoid false conclusions. For each paper, display ideas and controls are highlighted.

SECTION 1 - Ripley's K and Spatial Statistics Methods

Spatiopath: Statistical analysis of spatial patterns in tumor microenvironment images

Benimam et al. - *Nature Communications* 2025 | PMID: 40164621 | DOI: [10.1038/s41467-025-57943-y](https://doi.org/10.1038/s41467-025-57943-y)

Summary: Extends Ripley's K to analyze cell-cell AND cell-tumor boundary interactions in multiplexed IF images. Introduces a null-hypothesis framework (CSR permutation envelopes) to distinguish significant associations from random. Validated with synthetic simulations before application to lung tumor sections.

Your relevance: You could add a cell-to-tissue-boundary or cell-to-TJ-region proximity metric using the same extension of K. Their edge correction approach handles bounded tissue regions - critical for whole-ovary images.

Display ideas: K(r) curves with Monte Carlo confidence bands (ribbon). Heatmaps of significant vs. random spatial zones.

Controls used: Synthetic simulation validation before real data. CSR permutation envelope (n=1000 randomizations).

Benchmarking Spatial Co-Localization Methods for Single-Cell Multiplex Imaging

Soupir et al. - *Statistics and Data Science in Imaging* 2025 | PMID: 40051984 | DOI: [10.1080/29979676.2024.2437947](https://doi.org/10.1080/29979676.2024.2437947)

Summary: Head-to-head benchmarking of 6 spatial co-localization metrics across simulated and real cancer imaging data. Tests statistical power and type I error for each metric.

Your relevance: Key finding: Ripley's K and pair correlation g(r) have the greatest power to detect co-localization. Other commonly used metrics have poor power. Justifies your use of K and motivates adding g(r).

Display ideas: Power curves vs. effect size. ROC-style comparison of metrics.

Controls used: Simulated data with known spatial structure. Type I error checked with random data.

Tumor immune cell clustering and its association with survival in ovarian cancer

Wilson et al. - *PLoS Computational Biology* 2022 | PMID: 35235563 | DOI: [10.1371/journal.pcbi.1009900](https://doi.org/10.1371/journal.pcbi.1009900)

Summary: Ripley's K + permutation-based CSR test applied to multiplex IF data from 160 ovarian cancer patients. Compares cell abundance and spatial clustering separately; shows combining both metrics discriminates survival better than either alone.

Your relevance: Strong evidence that spatial clustering metrics add information beyond cell counts. The K-curve summary statistic (area between observed and CSR) converts the curve into a scalar for statistical comparison across conditions.

Display ideas: Per-sample K summary statistic as boxplots by condition. Kaplan-Meier curves stratified by spatial clustering.

Controls used: Permutation of cell labels within each tissue section (n=1000). Separate validation cohort (TMA).

spatialTIME and iTIME: R package for spatial analysis of immunofluorescence data

Creed et al. - *Bioinformatics* 2021 | PMID: 34734969 | DOI: [10.1093/bioinformatics/btab757](https://doi.org/10.1093/bioinformatics/btab757)

Summary: Open-source R package implementing Ripley's K, Besag's L, Macdonald's M, and G-function (nearest-neighbor CDF). Includes a Shiny web app for non-programmers. Produces publication-quality spatial organization plots.

Your relevance: Ready-to-use toolbox. The M function (normalized K) is scale-free and comparable across samples of different sizes. The L(r)-r plot centers CSR at zero, making deviations immediately visible.

Display ideas: L(r)-r curves (CSR baseline at zero). Univariate and bivariate spatial plots per sample.

Controls used: CSR permutation envelopes built into the package.

Geospatial distribution of cancer-associated fibroblasts in renal cell carcinoma

Chakiryan et al. - *Cancers* 2021 | PMID: 34359645 | DOI: [10.3390/cancers13153743](https://doi.org/10.3390/cancers13153743)

Summary: Uses normalized Ripley's K at a fixed radius (nK(25) at 25 um) as a single summary scalar per sample. Also computes median nearest-neighbor distance between two cell types (CAFs and proliferating tumor cells). Links spatial metrics to patient survival and therapy resistance.

Your relevance: The 'K at fixed radius' approach converts the K curve into one number per sample - directly usable in boxplots, ANOVAs, or regressions comparing experimental conditions. Choose radius based on biology (e.g., expected contact distance).

Display ideas: Boxplots of nK(r) and NN distance by condition. Scatter of NN distance vs. cell density.

Controls used: nK=1 represents CSR; test whether observed significantly differs. Multivariable Cox regression adjusting for covariates.

Analysis of spatial structure of epidermal nerve fibre patterns from replicated data

Myllymaki et al. - *Journal of Microscopy* 2012 | PMID: 22906010 | DOI: [10.1111/j.1365-2818.2012.03636.x](https://doi.org/10.1111/j.1365-2818.2012.03636.x)

Summary: CRITICAL METHODOLOGICAL PAPER. Presents a mixed model approach for Ripley's K applied to replicated spatial data (multiple images per subject/condition). Accounts for within-group vs. between-group variability.

Your relevance: When you have multiple ovary images per condition, you cannot pool K functions naively - images from the same condition are not fully independent. This paper shows how to correctly compare groups with replicates.

Display ideas: Pooled K function curves per group with confidence bands derived from mixed model.

Controls used: Mixed model F-test for group differences in K(r) across a range of radii.

Pairwise interaction Markov model for 3D epidermal nerve fibre endings

Konstantinou & Sarkka - *Journal of Microscopy* 2022 | PMID: 36106649 | DOI: [10.1111/jmi.13142](https://doi.org/10.1111/jmi.13142)

Summary: Applies anisotropic cylindrical Ripley's K for 3D confocal data where Z resolution differs from XY. Models spatial interactions in 3D with an anisotropy factor.

Your relevance: Your 3D confocal data has different pixel sizes in Z (0.25 μ m) vs XY (0.14 μ m). Your current Ripley's K implementation should use the anisotropy factor (1.78) - this paper shows the theory.

Display ideas: K-function in cylindrical coordinates. Comparison of healthy vs. diabetic neuropathy groups.

Controls used: Anisotropy factor validated by comparison with isotropic model on synthetic data.

SECTION 2 - Cell Contact Graphs and Network Analysis

Geometric and network organization of visceral organ epithelium

Liu et al. - *Frontiers in Network Physiology* 2023 | PMID: 37234691 | DOI: [10.3389/fnetp.2023.1144186](https://doi.org/10.3389/fnetp.2023.1144186)

Summary: Converts cell segmentation into a cell contact network (cell = node, shared border = edge). Computes polygon distributions, graphlet (subgraph) frequencies using EpiGraph software, and clustering coefficient. Compares heart, lung, liver, and bowel epithelia against mathematical reference patterns.

Your relevance: Graphlet frequency analysis is a richer description of local network topology than clustering coefficient alone. EpiGraph is open source and applicable to your label images. Directly extends your current graph analysis.

Display ideas: Radar plots comparing graphlet frequencies across conditions. Polygon distribution histograms.

Controls used: Three reference patterns: mathematical hexagonal grid, randomized, and Voronoi tessellation. Results compared against all three to determine which reference the observed tissue most resembles.

Evolving topological order in the postnatal visceral pleura

Liu et al. - *Developmental Dynamics* 2024 | PMID: 38169311 | DOI: [10.1002/dvdy.688](https://doi.org/10.1002/dvdy.688)

Summary: Applies Voronoi tessellation + graphlet motif frequencies + graph theory (clustering coefficient, network heterogeneity) to track developmental changes in tissue organization from postnatal day 1 to P15. Clustering coefficient decreases progressively - showing these metrics track biologically meaningful changes.

Your relevance: Template for comparing spatial topology across experimental time points or conditions. Shows that graph metrics can detect subtle progressive changes in tissue organization.

Display ideas: Line plots of clustering coefficient over developmental stages. Side-by-side polygon distribution bar charts.

Controls used: Developmental progression used as positive control (P1 vs P15 expected to differ).

Fundamental physical cellular constraints drive self-organization of tissues

Sanchez-Gutierrez et al. - *EMBO Journal* 2015 | PMID: 26598531 | DOI: [10.15252/embj.201592374](https://doi.org/10.15252/embj.201592374)

Summary: Uses Voronoi tessellation to characterize cell packing across diverse tissues including *Drosophila*. Shows polygon frequency distributions correlate with cell area distributions across species. Challenges the assumption of a universal stereotyped polygon distribution.

Your relevance: Directly uses *Drosophila* tissue and demonstrates Voronoi analysis reveals differences between normal and perturbed tissue states (disease, genetic). Voronoi tessellation could complement your face-adjacency contact graph.

Display ideas: Polygon type frequency distributions (bar charts). Scatter of polygon frequency vs. cell area CV.

Controls used: Computer model perturbations of force balance. Comparison across disease/genetic conditions and species.

SECTION 3 - 3D Segmentation and Spatial Phenotyping Pipelines

ShapeMetrics: A user-friendly pipeline for 3D cell segmentation and spatial tissue analysis

Takko et al. - *Developmental Biology* 2020 / Pajanoja & Kerosuo - *Methods Mol Biol* 2024 | PMID: 32061886 / 37219813 | DOI: [10.1016/j.ydbio.2020.02.003](https://doi.org/10.1016/j.ydbio.2020.02.003)

Summary: MATLAB pipeline: Ilastik-based segmentation -> 3D morphometrics (volume, surface area, ellipticity, elongation, longest axis) -> unsupervised heatmap clustering by morphometric features -> spatial mapping of clusters back to tissue.

Your relevance: The spatial remapping step is powerful: it reveals whether morphometrically distinct cell subpopulations are spatially clustered. You could apply this to your VASA cell morphometrics to ask whether larger/smaller cells are located in specific ovary regions.

Display ideas: Heatmaps of morphometric clusters. 3D spatial maps color-coded by cluster membership. Volume/surface scatter plots.

Controls used: Unsupervised clustering used without prior assumptions about subgroups.

Integrated Cytometry With Machine Learning for human kidney tissue (VTEA)

Winfree et al. - *Laboratory Investigation* 2023 | PMID: 36867975 | DOI: [10.1016/j.labinv.2023.100104](https://doi.org/10.1016/j.labinv.2023.100104)

Summary: VTEA (Volumetric Tissue Exploration and Analysis): integrates 3D segmentation, interactive cytometry, machine learning cell classification, and neighborhood analysis for multiplexed confocal images of human kidney.

Your relevance: The neighborhood analysis module computes cell-type enrichment in local neighborhoods. Applicable to asking: do VASA cells near TJ cells differ morphometrically from isolated VASA cells?

Display ideas: Interactive scatter plots (cytometry-style). Spatial maps of cell subtypes and neighborhoods.

Controls used: Machine learning cross-validation for cell classification accuracy.

EASI-FISH for thick tissue defines lateral hypothalamus spatio-molecular organization

Wang et al. - *Cell* 2021 | PMID: 34875226 | DOI: [10.1016/j.cell.2021.11.024](https://doi.org/10.1016/j.cell.2021.11.024)

Summary: Expansion microscopy + StarDist 3D segmentation + spatial mapping of dozens of molecularly defined cell types in 300 um thick brain sections. Identifies 9 spatially defined subregions invisible to traditional anatomy.

Your relevance: Demonstrates that 3D spatial analysis of segmented cell types can reveal tissue subdomains not visible by anatomy alone. Inspires asking whether VASA cell populations have spatially distinct subregions within the ovary.

Display ideas: 3D spatial maps with cell types color-coded. Cluster centroid heatmaps across tissue volume.

Controls used: scRNA-seq cross-validation of spatially-defined clusters. Iterative marker refinement.

SECTION 4 - Directly Relevant Biology (Drosophila Ovary)

Specification and spatial arrangement of cells in the germline stem cell niche of the Drosophila ovary

Panchal et al. - *PLoS Genetics* 2017 | PMID: 28542174 | DOI: [10.1371/journal.pgen.1006790](https://doi.org/10.1371/journal.pgen.1006790)

Summary: Studies Traffic jam (Tj) transcription factor - the same TJ marker used in the pipeline - and its role in determining spatial organization of germline stem cell niche cells (cap, escort, terminal filament cells). Shows Tj controls niche architecture and stem cell carrying capacity (normally 2-3 GSCs per niche).

Your relevance: Direct biological context for your TJ channel. Establishes that TJ-expressing cells' spatial arrangement is functionally significant. Provides baseline values for GSC number and niche cell organization in wildtype.

Display ideas: Confocal images of niche architecture. Cell count quantifications per niche. Genetic perturbation comparisons.

Controls used: Loss-of-function and partial loss-of-function compared to wildtype. Notch pathway interaction as control.

SECTION 5 - Summary: New Metrics to Consider

The following metrics are used in the literature reviewed and could be added to your spatial analysis pipeline. All are applicable to segmented label images with known physical pixel sizes.

Metric	Key Paper	What it adds vs. current pipeline
Pair correlation function $g(r)$	Soupir 2025	Derivative of $K(r)$; shows clustering at specific spatial scales rather than cumulatively. Higher statistical power than K alone.
M function (normalized K)	Creed 2021 (spatialTIME)	Scale-free version of K . Comparable across samples of different sizes - critical when ovaries vary in volume.
K at fixed radius (scalar summary)	Chakiryan 2021	Single number per sample from K curve. Enables boxplots, ANOVA, regression across conditions without curve comparison.
NN distance between two cell types	Chakiryan 2021	Direct biological proximity: e.g., median distance from each VASA cell to the nearest TJ cell.
Graphlet frequency (EpiGraph)	Liu 2023/2024	Richer local topology descriptor than clustering coefficient alone. Detects subtle topological differences.
Polygon distribution	Sanchez-Gutierrez 2015, Liu 2023	Cell packing topology from Voronoi. Compare to hexagonal/random/Voronoi references.
Morphometric cluster + spatial remapping	Takko 2020 (ShapeMetrics)	Reveals spatial gradients in cell shape - are larger VASA cells located in specific ovary regions?
Cell-to-boundary proximity	Benimam 2025 (SpatioPath)	Distance of VASA cells from tissue boundary or from TJ-labeled region. Adds structural context.

SECTION 6 - Critical Controls: What the Literature Recommends

1. Cell density as covariate

Higher cell density mechanically forces shorter nearest-neighbor distances. Always report density alongside spatial metrics and test whether differences in spatial patterns remain after controlling for density. (Chakiryan 2021, Soupir 2025)

2. Permutation test for Ripley's K

Randomly shuffle cell positions (or cell type labels) within each sample $n=1000$ times to build a null distribution. Your current pipeline uses the configuration model for graph topology - extend this to K as well. (Wilson 2022, Benimam 2025)

3. Edge correction for bounded tissues

Cells near the tissue boundary have fewer potential neighbors, artificially reducing K . Apply isotropic or Ripley's border correction. Critical when ovaries vary in shape and size. (Benimam 2025, Konstantinou 2022)

4. Mixed model for replicated data

When comparing conditions with multiple images per condition, treat each image as one replicate. Do not pool all images from a condition - this ignores within-condition biological variability. Summarize each image to a scalar (e.g., K at fixed radius) then compare scalars across conditions. (Myllymaki 2012)

5. Type I error validation

Before applying to real data, run your spatial metrics on simulated random point patterns. Verify that the test correctly does not reject CSR when data are actually random. (Soupir 2025, Benimam 2025)

6. Biological positive control

A genetic or pharmacological perturbation known to disrupt cell organization should produce a detectable change in your spatial metrics. If it does not, your metrics may lack sensitivity. (Panchal 2017, Sanchez-Gutierrez 2015)

7. Reference pattern comparison for graphs

When reporting clustering coefficient or graphlet frequencies, always compare to at least two reference patterns: (a) a fully random graph with the same degree sequence (configuration model - you already do this), and (b) a Voronoi tessellation of randomly placed points in the same volume. (Liu 2023, Sanchez-Gutierrez 2015)

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